

DataHub1000 Remote Setup



Background

SolaX DataHub1000 is the professional device for monitoring and managing sites with multiple SolaX inverters, more and more customers are now using DataHub1000 to manage their sites.

However, it is not convenient to set up on the local page, as it may be necessary to travel to the site from a long distance. So we launched the DataHub1000 remote setup function which greatly saves your time for system configuration and troubleshooting.

Where to remote setup?

Step 1

Open the Databus page in Device in the left navigation bar of the cloud platform, then click the Registration Number.

The screenshot shows the SolaX Power cloud platform interface. On the left is a dark navigation bar with icons and labels for Overview, Agents, Users, System&Site, Device, Inverter, EV Charger, DataHub, and Alarm Management. The 'Device' menu item is highlighted with a red box and labeled '1'. Below it, the 'DataHub' sub-menu is also highlighted with a red box and labeled '2'. The main content area displays a table of devices. The first row of the table has the registration number 'SH6JYTQ3VJ' highlighted with a red box and labeled '3'. Above the table, there are search filters for Site Name, Registration No. (SH6JYTQ3VJ), and Online Status. The table has columns for No., Registration No., Number of meters, Site Name, Device Type, Serial port 1 (Number of devices, Number of devices online), Serial port 2 (Number of devices, Number of devices online), and Device Type. The first row of data shows an Inverter with 12 devices in serial port 1 and 0 in serial port 2, and a Meter in serial port 2. At the bottom right of the table, there is a pagination control showing 'Total 1', '10/page', and 'Go to 1'.

No.	Registration No.	Number of meters	Site Name	Device Type	Serial port 1 Number of devices	Number of devices online	Serial port 2 Number of devices	Number of devices online	Device Type
1	SH6JYTQ3VJ	2	BCC***500	Inverter	12	0	0	0	Meter

Step 2

Open the remote setup interface, showing as follows.

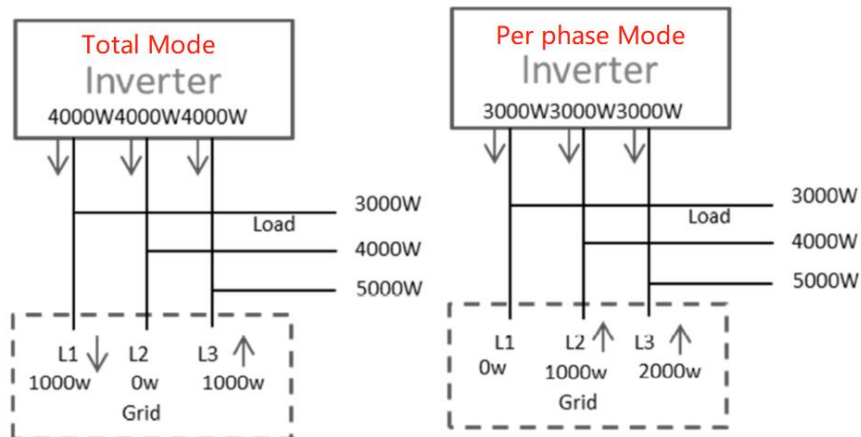
SOLAX POWER							
Registration							
No. SH6JY							
Device Details							
Export Limit Control							
Power Control							
Other Control							
Internet Setting							
Time Setting							
Serial Port Setting							
Other Setting							
Datahub Info							
RS485 Channel	Device ID	SN	Device Type	Operating Status	Inverter Error Number	Version	
1	1	MPT30R6204030	INV	unknown	0	A1.20D1.22	
1	2	MPT30R6204028	INV	unknown	0	A1.20D1.22	
1	3	MPT30R6204025	INV	unknown	0	A1.20D1.22	
1	4	MPT30R6204022	INV	unknown	0	A1.20D1.22	
1	5	MPT30R6204005	INV	unknown	0	A1.20D1.22	
1	6	MPT30R6204009	INV	unknown	0	A1.20D1.22	
1	7	MPT30R6204019	INV	unknown	0	A1.20D1.22	
1	8	MPT30R6204013	INV	unknown	0	A1.20D1.22	
1	9	MPT30R6204011	INV	unknown	0	A1.20D1.22	
1	10	MPT30R6204014	INV	unknown	0	A1.12D1.15	
1	11	MPT30R6204023	INV	unknown	0	A1.20D1.22	
1	12	MPT30R6204021	INV	unknown	0	A1.20D1.22	
3	2	meter-2	ELM	online	0	0.00	
4	1	meter-1	ELM	online	0	0.00	

What remote setup items are available?

① Export Limit Control

A. Mode selection for feed-in power control - 'Total' or 'Per phase'

The screenshot shows the 'Export Limit Control' settings page. The 'Control Mode' dropdown is set to 'Total'. The 'Feed Redundancy' is set to 'Total'. The 'Export Power (%)' is set to 0. The 'Reset' and 'Submit' buttons are visible at the bottom right.



Note: For three-phase three-wire systems, per phase mode is not available

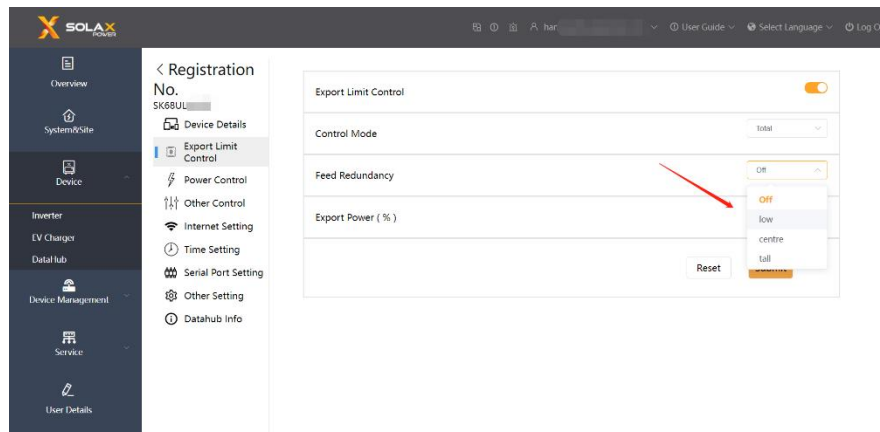
B. Feed Redundancy

Off: 1% -- Indicates that when the load is stabilized, the feed-in power fluctuates between $\pm 1\%$. The feed-in energy may not be completely 0, but the inverter will take less energy from grid.

low: 2.5% -- The power taken from the grid = Total rated power of the site * 2.5%.

centre: 4.5% -- The power taken from the grid = Total rated power of the site * 4.5%.

tall: 6.5% -- The power taken from the grid = Total rated power of the site * 6.5%.



Note: The higher Feed Redundancy threshold, the better the zero control, but less power will be generated accordingly.

C. Export Power

The system feed-in power is set as percentage with an accuracy of 0.1%.

Export Limit Control	☒
Control Mode	Total
Feed Redundancy	
Export Power (%)	0
Reset Submit	

② Power Control

Customers can use external power controllers to control the power output of the system and work with the grid to accomplish load management.

A. Ripple Control Receiver

Users can define output settings for 16 states which are controlled according to DI port high and low input states.

Setting items include: inverter output active power control, inverter input active power

control, inverter output reactive power control and inverter shutdown.

☒ Ripple Control Receiver
 ☐ DRED Control
 ☐ Disable

D1 D2 D3 D4	Status	Setting	Active Power(0~100%)	Power Factor(80 ~ 100%)	Reactive Mode
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼
● ● ● ●	●	▼	0 ~ 100	80 ~ 100	▼

B. DRED control (Mandatory under Australian safety)

There are 6 DRM control modes here according to Australian safety regulations.

☐ Ripple Control Receiver
 ☒ DRED Control
 ☐ Disable

Current Status	DRM Mode	A0	A1	D1	D2	D3	Reactive Mode
💡	0	●	●	●	●	●	Turn Off
💡	1	●	●	●	●	●	Input active power 0%
💡	5	●	●	●	●	●	Output active power 0%
💡	6	●	●	●	●	●	Output active power 50%
💡	7	●	●	●	●	●	Output active power 75%
💡	8	●	●	●	●	●	Output active power 100%

Submit

③ Other Control

A. Meter Reversion

You can observe the **Pgrid** data on the inverter LCD screen or the **DataHub Meter** data in the platform site page to analyse whether the feed-in data is reversed.

Site Analysis Device Matrix Statistic Report **DataHub Meter**

Site Name: XXXXXXXXXX Monitoring data refresh interval: 5 min

Time Horizon: 2023-07-18 - 2023-08-18 Registration No. String

No.	Registration No.	String	Grid Power(W)	Feed-in Energy(kWh)	Consume Energy(kWh)	Update Time
1	XXXXXXXXXX	meter-1	34392	100961.7	85133.1	2023-08-18 07:33:58
2	XXXXXXXXXX	meter-1	29436	100959.1	85133.1	2023-08-18 07:28:56
3	XXXXXXXXXX	meter-1	26052	100956.9	85133.1	2023-08-18 07:23:57
4	XXXXXXXXXX	meter-1	22584	100954.8	85133.1	2023-08-18 07:18:56
5	XXXXXXXXXX	meter-1	19908	100953.1	85133.1	2023-08-18 07:13:57
6	XXXXXXXXXX	meter-1	14088	100951.6	85133.1	2023-08-18 07:08:55
7	XXXXXXXXXX	meter-1	12456	100950.7	85133.1	2023-08-18 07:03:56
8	XXXXXXXXXX	meter-1	4080	100949.9	85133.1	2023-08-18 06:58:55
9	XXXXXXXXXX	meter-1	4680	100949.5	85133.1	2023-08-18 06:53:55
10	XXXXXXXXXX	meter-1	2112	100949.3	85133.1	2023-08-18 06:48:57

Total 8986 10/page < 1 2 3 4 5 6 ... 899 > Go to 1

If the data is reversed, you can correct it by setting the Meter Reversion on this page.

Meter Reversion

meter-2	☐
meter-1	☐

ResetSubmit

B. Main Breaker Limit

If the DataHub has EV-Charger connected, user can set the Main Breaker Limit current here. The DataHub can compare the real-time system's GCP (Grid Connection Point) current with this value to dynamically control the EV-Charger current, so the system value will not exceed the limitation of the main breaker.

Note: If the system has multiple EV-Charger connected, DataHub will divide the charge current of each EV-Charger equally.

Main Breaker Limit

Main Breaker Limit	60
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ResetSubmit

C. Max. Unbalance Power

If the system has single-phase EV-Charger connected, while the local grid does not allow power unbalance, then customer can set Max Unbalance Power here to limit the three-phase unbalance power.

Max. Unbalance Power

Max. Unbalance Power	4.6KW
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ResetSubmit

④ Settings for Network, Time, and Serial Port

Note: The Baud Rate setting of the DataHub serial port needs to be consistent with the baud rate of the device connected to this port.

Please refer to the following table, for device baud rate:

Device	Baud Rate
X1-MINI G4	9600
X1-BOOST G4	9600
X3-MIC G2	9600
X3-PRO G2	9600
X3-MEGA G2	9600
X3-FORTH	9600
X1-Hybrid G4	19200
X3-Hybrid G4	19200
EV-Charger	9600
SolaX Meter	9600

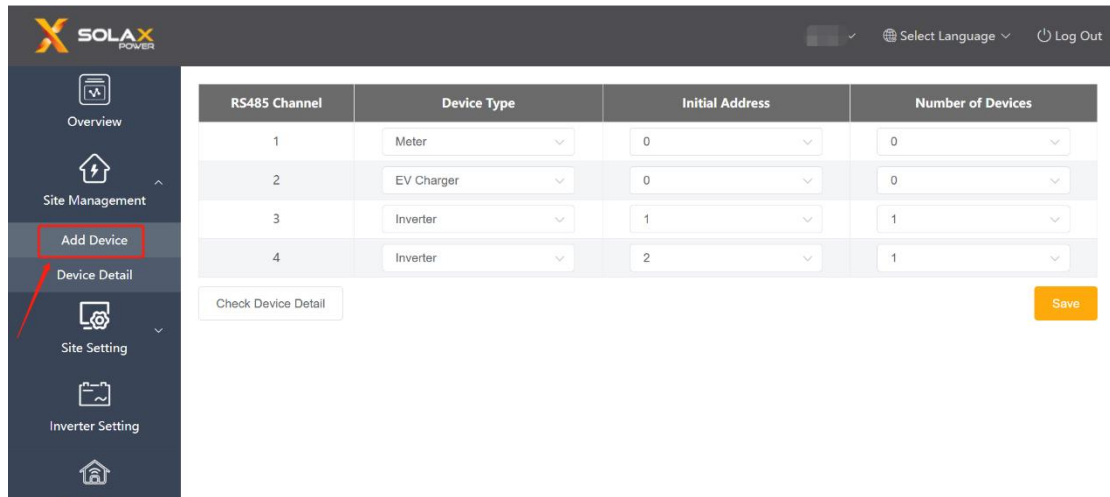
What are the differences between the local setup and the SolaX

Cloud remote setup?

The remote settings are more simplified than local settings, some settings are only reflected in local.

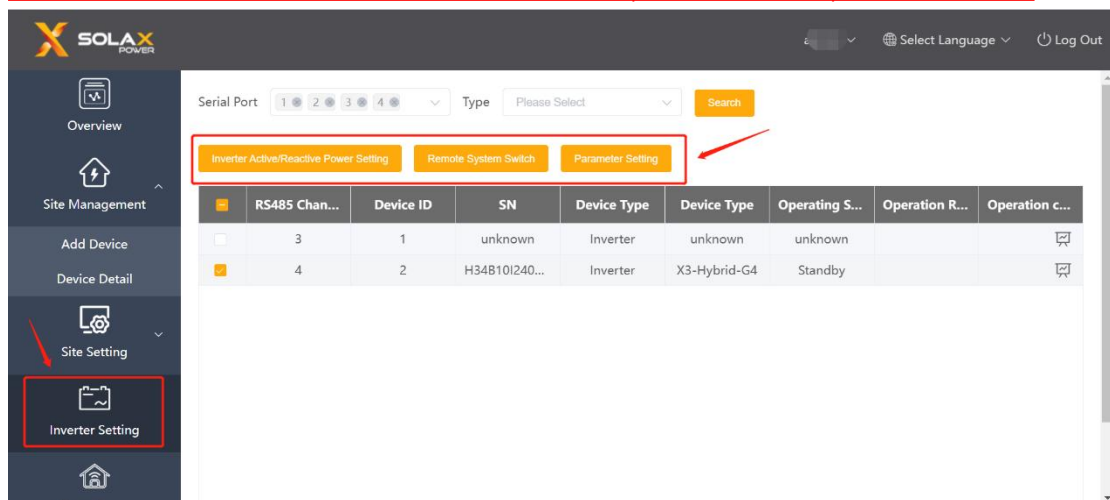
- **Add Device:** Add inverter, meter, EV-Charger and weather station, set their starting address and number of devices;

Note: The address setting on the DataHub need consist with the device address to ensure the communication between the DataHub and the device.

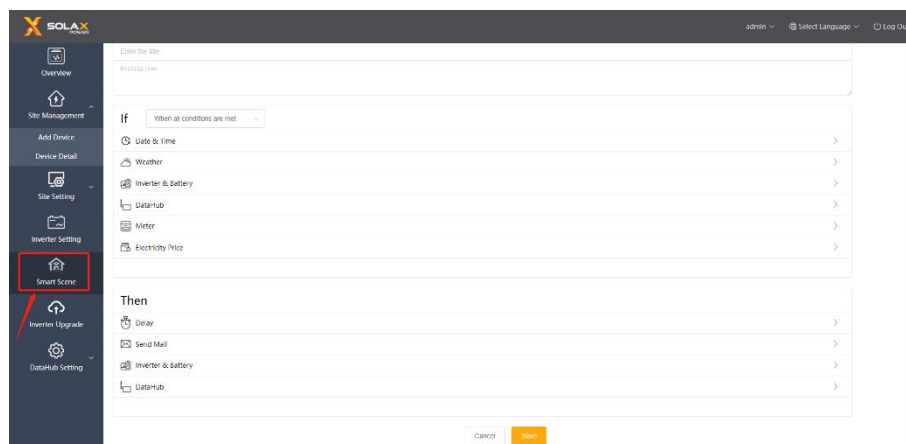


- **Inverter Setting:** You can set the Active/Reactive Power, Remote System Switch, IV curve scanning and Modbus TCP read/write operation here.

Note: The remote on/off function here **does not** require an external power controller.



- **Smart Scene:** Customers can customize various rules here to automate and smarten their home PV system. For example, charge the battery when the electrical price falls below a certain value.



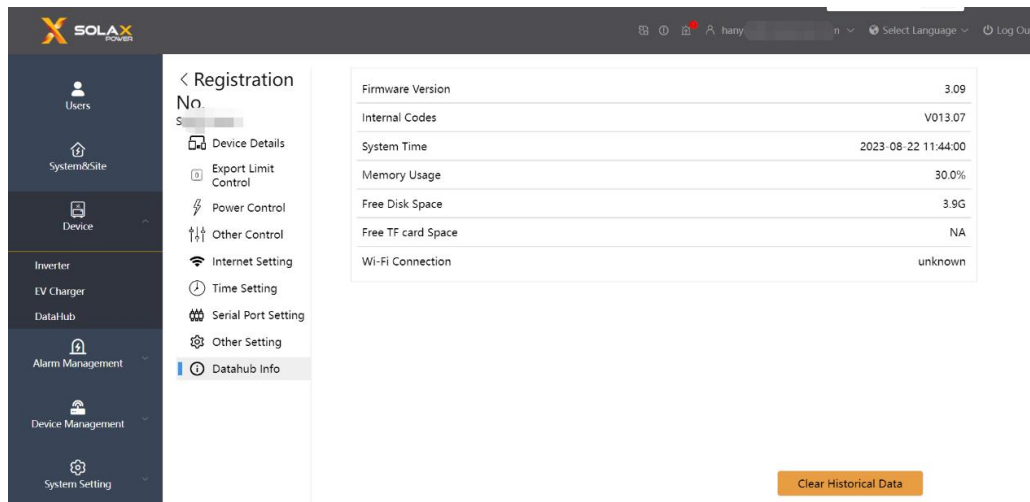
Version requirements

The remote setup function requires the following internal code for DataHub:

Datahub Internal Codes ≥ 12.07

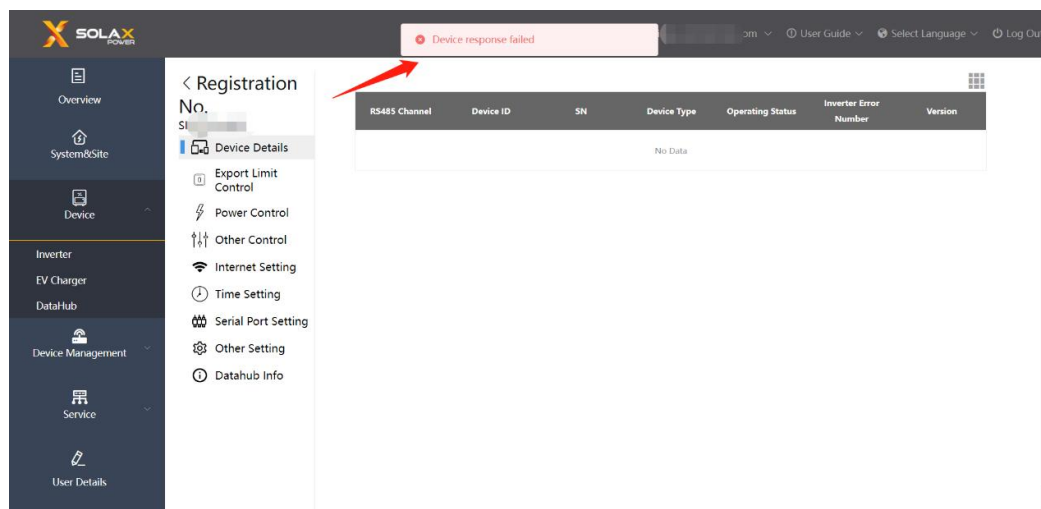
You can confirm the internal code of DataHub from SolaX Cloud:

Datahub Setting → Datahub Info → Internal Codes



Parameter	Value
Firmware Version	3.09
Internal Codes	V013.07
System Time	2023-08-22 11:44:00
Memory Usage	30.0%
Free Disk Space	3.9G
Free TF card Space	NA
Wi-Fi Connection	unknown

If the internal code does not meet the condition, the remote setup will not be successful.



RS485 Channel	Device ID	SN	Device Type	Operating Status	Inverter Error Number	Version
No Data						